Prasad Patil

Technical Architect | Research Scholar | Cloud, AI & Microservices Expert

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Summary

- Experienced Research and Technology Professional with over 14 years in software development, applied research, and cloud transformation, currently pursuing a PhD in Quantum Computing, Cryptography, and Machine Learning.
- Proficient in GenAI-based legacy code analysis and migration, microservices architecture, and cloud-native design, with a deep focus on Microsoft Azure solutions and AI-driven transformations.
- Blend of academic rigor and industrial experience, with strong analytical, problem-solving, and technical writing skills. Published patents and a paper on intelligent compliance systems.
- Well-versed in prompt engineering and AI tools; recognized for contextual innovations in mainframe modernization using GenAI.

Experience

08/2023–Present **Technical Architect, MBU Apps Infra Engg. - AI.cloud Unit**, *Tata Consultancy Services (TCS)*, Nagpur, India
- Led research and implementation initiatives for Azure Cloud Migration lifecycle tools, integrating offerings like Cloud Console, Migration Factory Hub, and Azure Well-Architected Framework.
- Architected solutions using GenAI for COBOL to Java migration with contextual understanding of legacy systems.
- Designed modular, microservices-based systems following Domain-Driven Design (DDD) principles.

04/2019–07/2023 **Technical Architect, ISN - Corporate Unit**, *TCS*, Pune, India
- Core contributor to ISN (Intelligent Services Networks) with focus on Spring Boot microservices and Azure cloud integration.
- Transformed legacy applications to microservices using DDD; deployed robust enterprise solutions on Azure.

04/2014–03/2019 **Full Stack Developer & Researcher, Ignio Project - CTO & Digitate Unit**, *Tata Research Development and Design Centre (TRDDC)*, Pune, India
- Specialized in compliance automation for networks and operating systems; developed scalable full-stack solutions.
- Contributed to policy definition, compliance engines, and integrations for diverse enterprise infrastructure.

03/2011–03/2014 **Developer & Researcher, eTransform Project - CTO Unit**, *TRDDC*, Pune, India
- Built standalone Java Swing modules for migration and transformation estimations.
- Leveraged multithreading, design patterns, and data structures to develop performance-sensitive applications.

Education

Present **PhD Scholar in Quantum Computing, Cryptography & Machine Learning**, *IIIT Nagpur*, Nagpur, India
Recipient of Fellowship from IIT Patna

2010 **B.E. in Computer Science and Engineering**, *Shri Ramdeobaba Kamla Nehru Engineering College*, Nagpur, India
2nd prize in internal coding competition

2007 **Diploma in Computer Technology**, *Nagpur Polytechnic*, Nagpur, India
Gold medalist for 3 consecutive years

Skills

- Technical Research, Microservices Architecture, GenAI, Prompt Engineering, Legacy Modernization, Advanced Java, Spring Boot, Cloud Architecture

Certifications

- NPTEL Certified: Quantum Algorithms and Cryptography
- NPTEL Certified: Machine Learning for Engineering and Science Applications
- Microsoft Certified: Azure Solutions Architect Expert (AZ-305)
- Microsoft Certified: Azure Administrator Associate (AZ-104)

Patents and Publications

- Submitted Paper: Hybrid Quantum DOA Algorithm to IEEE TIM (2025)
- Indian Patent: Compliance as a Service (2024)
- US Patent: Compliance Automation Platform (2017)
- Technical Paper: Compliance as a Service (2017)

Awards and Honors

- Contextual Master: GenAI for Mainframe Modernization (2024)
- Best Performer of the Year, TCS (2021)
- 2nd Rank, Ignio Hackathon (2019)
- Special Mention, Data Privacy Day Challenge Fest (2014)
- 1st Place, Initial Learning Program Training, TCS (2011)